

Inception Report

*AI-Generated Image Detection for Secure Liveness
Verification in Pathao Pay*

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1 Complex Engineering Problem Statement

The development of an AI-Generated Fraud Detection system for Secure Liveness and KYC Verification in Pathao Pay presents a multifaceted technical challenge that addresses critical security vulnerabilities in modern financial technology platforms. As digital financial services expand rapidly in emerging markets, the sophistication of fraud attempts has evolved in parallel, with adversaries increasingly leveraging generative AI technologies to circumvent traditional verification mechanisms. This project tackles the intersection of computer vision, machine learning security, real-time systems optimization, and regulatory compliance within the constraints of resource-limited mobile environments.

1.1 AI-Generated Fraud Detection

The primary challenge involves identifying increasingly realistic AI-generated and manipulated facial images used to bypass financial liveness verification systems. Modern generative adversarial networks (GANs), diffusion models, and deepfake technologies can produce synthetic faces that exhibit photorealistic qualities, including natural skin textures, lighting variations, and even simulated micro-expressions. The detection system must discriminate between genuine live captures and sophisticated forgeries that may include:

- High-fidelity GAN-generated synthetic identities with consistent facial features across multiple angles
- Deepfake face-swap attacks replacing genuine users with synthetic or stolen identities
- Neural rendering techniques that introduce realistic eye reflections, skin imperfections, and environmental lighting
- Hybrid attacks combining genuine photographs with AI-enhanced manipulations to defeat forensic analysis
- Adversarial perturbations specifically designed to fool liveness detection classifiers

The system must operate as a discriminator in an adversarial environment where attackers continuously probe detection boundaries and adapt their techniques based on observed failures. This requires developing robust feature extraction methods that identify subtle statistical anomalies, frequency domain artifacts, and neural network fingerprints that persist even in state-of-the-art generative models.

1.2 Artifact Evolution Resistance

Designing detection methods that remain effective as generative models continuously reduce visual and statistical artifacts represents a fundamental arms race problem. Each iteration of generative models addresses previous detection mechanisms by:

- Reducing frequency domain inconsistencies that traditional forensic methods exploit
- Improving temporal coherence in video-based deepfakes to eliminate frame-to-frame artifacts

- Eliminating checkerboard patterns, boundary artifacts, and color bleeding characteristic of earlier GAN architectures
- Normalizing statistical distributions of pixel intensities to match natural image statistics
- Incorporating physics-based rendering to generate plausible eye reflections and lighting interactions

The detection system must employ ensemble approaches that combine multiple orthogonal detection strategies, including behavioral biometrics (natural head movement patterns), physical challenge-response protocols (random prompt compliance), and deep learned representations that capture imperceptible generative artifacts. Furthermore, the system must support continuous learning and model updates to adapt to emerging generative techniques without requiring full system redeployment.

1.3 Precision versus User Experience Trade-off

Balancing high fraud detection accuracy with low false positive rates to avoid blocking legitimate users during KYC processes presents a critical operational constraint. Financial services in competitive markets cannot afford user friction that drives customer abandonment, yet security compromises can result in regulatory violations, financial losses, and reputational damage. The challenge encompasses:

- Establishing optimal decision thresholds that minimize Type I errors (false positives) while maintaining acceptable Type II error rates (false negatives)
- Implementing graceful degradation strategies where marginal cases receive additional verification steps rather than immediate rejection
- Designing intuitive user guidance systems that help legitimate users successfully complete verification without revealing exploitable information to attackers
- Developing confidence scoring mechanisms that enable human-in-the-loop review for ambiguous cases

Statistical modeling must account for population diversity, including variations in skin tones, facial features, age-related characteristics, and accessibility needs for users with physical disabilities. The system must achieve consistent performance across demographic segments to prevent algorithmic bias while maintaining security effectiveness.

1.4 Real-World Robustness

Ensuring detection performance remains stable under image compression, resizing, camera noise, and adversarial manipulation requires extensive robustness engineering. Real-world deployment scenarios introduce numerous confounding factors:

- Diverse mobile camera hardware with varying sensor qualities, lens characteristics, and image processing pipelines
- Inconsistent lighting conditions ranging from direct sunlight to low-light indoor environments with artificial illumination

- Network-induced JPEG compression artifacts and resolution downsampling that may obscure subtle forensic features
- Environmental factors including background clutter, motion blur, and partial occlusions
- Adversarial attacks specifically designed to exploit model vulnerabilities through imperceptible perturbations
- Presentation attacks using printed photographs, digital displays, 3D masks, or video replay

The detection pipeline must incorporate preprocessing normalization, multi-scale analysis to maintain sensitivity across resolution variations, and adversarial training techniques that expose models to anticipated attack vectors during development. Robustness validation requires extensive testing across device profiles, environmental conditions, and synthetic attack datasets.

1.5 Lightweight Real-Time Performance

Optimizing models to meet strict latency and resource constraints of mobile-first financial applications presents significant architectural challenges. The system must operate within:

- Latency budgets of under 2-3 seconds for complete verification to maintain acceptable user experience
- Memory footprints in Megabytes scale to accommodate low-end Android devices common in emerging markets
- CPU-only inference capabilities for devices lacking GPU acceleration
- Power efficiency constraints to prevent battery drain during verification processes

Achieving real-time performance necessitates model compression techniques including knowledge distillation, quantization-aware training, neural architecture search for efficient topologies, and hybrid edge-cloud architectures that perform lightweight screening on-device with selective cloud-based deep analysis for suspicious cases.

1.6 KYC and NID Consistency Verification

Building a robust KYC pipeline that cross-verifies live selfie data with National ID (NID) information while handling mismatches and edge cases securely represents a complex identity verification workflow. The system must:

- Perform accurate face matching between live selfie captures and NID photographs despite age progression, appearance changes (hairstyles, facial hair, accessories), and photograph quality differences
- Cross-reference biometric data with government databases while maintaining privacy compliance and data protection regulations

- Handle legitimate mismatch scenarios including name variations, expired documents, and temporary appearance changes due to medical conditions
- Implement secure document verification that detects forged, altered, or stolen identity documents
- Manage edge cases including first-time users without previous verification history, users with multiple valid identity documents, and account recovery scenarios

The verification workflow must support both synchronous real-time verification for immediate account access and asynchronous review processes for complex cases, while maintaining consistent security standards across both pathways.

1.7 Privacy-Preserving Architecture

Implementing liveness detection while respecting user privacy and complying with data protection regulations (GDPR, local privacy laws) requires careful architectural design:

- Minimizing biometric data retention and implementing secure deletion protocols
- Employing on-device processing where feasible to prevent transmission of raw biometric data
- Implementing differential privacy techniques when aggregating detection patterns for model improvement
- Providing transparent disclosure mechanisms and user consent workflows
- Supporting data portability and user rights to access, correct, or delete verification data

1.8 Scalability and High-Availability Requirements

The Liveness Detection system deployed in the cloud must support large-scale deployment serving millions of verification requests with enterprise-grade reliability:

- Horizontal scalability to handle peak verification loads during promotional campaigns or market expansion
- Fault tolerance and graceful degradation when individual components fail
- Seamless deployment for model updates
- Comprehensive monitoring and observability for detecting performance degradation and emerging attack patterns

1.9 Legacy System Interoperability

Integrating advanced liveness detection into existing Pathao Pay infrastructure requires careful architectural planning:

- Backward compatibility with existing user accounts and verification records
- Gradual migration strategies that maintain service continuity
- API versioning and abstraction to support iterative system improvements
- Integration with existing fraud detection rules engines

1.10 Continuous Improvement Pipeline

Establishing data collection, model retraining, and deployment pipelines to maintain effectiveness against evolving threats:

- Active learning systems that identify challenging cases for manual review and dataset enrichment
- Threat intelligence integration to proactively adapt to emerging generative techniques
- Federated learning approaches that improve models while preserving user privacy

1.11 Reliable Evaluation Metrics

The system's effectiveness will be measured across multiple dimensions:

- **Security Metrics:** False acceptance rate (FAR), false rejection rate (FRR), presentation attack detection rate (PADR), and detection performance against state-of-the-art deepfake datasets
- **Performance Metrics:** End-to-end verification latency (p50, p95, p99), on-device inference time, memory footprint, and power consumption
- **User Experience Metrics:** Verification success rate on first attempt, user abandonment rate during KYC, and customer support ticket volume related to verification issues
- **Operational Metrics:** System availability, scalability under load, model update deployment frequency, and incident response time

2 Roles and Work Distribution

2.1 Overview

The complex engineering problem of implementing AI-Generated Fraud Detection for Secure Liveness and KYC Verification in Pathao Pay requires a multidisciplinary approach. The team of five members has been strategically organized to address different technical domains while ensuring collaborative integration. Each member has primary ownership of specific components while contributing to overall system integration and testing.

2.2 Team Member Roles and Responsibilities

2.2.1 Tamzeed Mahfuz: AI Model Development and Deepfake Detection Lead

Primary Responsibilities:

- Design and develop the core liveness detection model capable of identifying AI-generated and manipulated facial images
- Research and implement detection techniques for various attack vectors (GANs, diffusion models, deepfakes, face swaps)
- Create and maintain training pipelines for continuous model improvement against evolving generative techniques
- Implement adversarial training strategies to improve model robustness

Problems to Solve:

- AI-Generated Fraud Detection: Identifying increasingly realistic AI-generated facial images
- Artifact Evolution Resistance: Maintaining effectiveness as generative models improve and reduce detectable artifacts
- Building ensemble detection methods that combine multiple orthogonal approaches for robust fraud identification

Key Deliverables:

- Trained liveness detection model with documented performance metrics
- Deepfake detection module with accuracy reports across various attack types
- Model versioning and continuous learning pipeline
- Technical documentation of detection methodologies

2.2.2 Nafis Nahian: Mobile Optimization and Real-Time Performance Lead

Primary Responsibilities:

- Optimize AI models for deployment on resource-constrained mobile devices
- Implement model compression techniques (quantization, pruning, knowledge distillation)
- Design hybrid edge-cloud architecture for efficient processing distribution
- Develop on-device inference pipeline with minimal latency and memory footprint
- Profile and optimize performance across various Android device specifications

Problems to Solve:

- Lightweight Real-Time Performance: Meeting strict latency constraints (under 2-3 seconds) on low-end devices
- Balancing model accuracy with computational efficiency for mobile-first applications
- Ensuring consistent performance across diverse hardware capabilities common in Bangladesh market
- Minimizing battery consumption during verification processes

Key Deliverables:

- Optimized mobile-compatible model with size in Megabyte scale
- Performance benchmarks across different device tiers
- Edge-cloud processing strategy with fallback mechanisms
- Battery and memory usage analysis reports

2.2.3 Ruwad Naswan: Robustness Engineering and Adversarial Defense Lead

Primary Responsibilities:

- Ensure system robustness under real-world conditions (compression, noise, varying lighting)
- Develop preprocessing and augmentation strategies for handling image quality variations
- Design and implement defenses against adversarial attacks and presentation attacks
- Create comprehensive test datasets covering edge cases and attack scenarios
- Implement image quality assessment and adaptive processing pipelines

Problems to Solve:

- Real-World Robustness: Maintaining stable performance under image compression, resizing, camera noise, and varying environmental conditions
- Defending against adversarial manipulation, presentation attacks (masks, displays, photos), and video replay attacks
- Handling diverse camera hardware quality and lighting scenarios prevalent in Bangladesh

Key Deliverables:

- Robustness testing framework with comprehensive attack datasets
- Image preprocessing and normalization pipeline
- Adversarial defense mechanisms with documented effectiveness
- Environmental condition handling strategies (lighting adaptation, noise reduction)

2.2.4 Aritra Debnath: KYC Pipeline and Identity Verification Lead**Primary Responsibilities:**

- Design end-to-end KYC verification workflow integrating liveness detection with NID verification
- Implement face matching algorithms between live selfies and NID photographs
- Create mismatch handling logic and edge case resolution protocols
- Integrate with government databases and third-party verification services

Problems to Solve:

- KYC & NID Consistency Verification: Building robust cross-verification between live selfie data and National ID information
- Handling legitimate mismatches (age progression, appearance changes, document quality differences)
- Managing edge cases (expired documents, name variations, multiple IDs, account recovery scenarios)
- Ensuring secure document extraction and validation while maintaining user privacy

Key Deliverables:

- Complete KYC verification pipeline with workflow documentation
- Face matching module with accuracy metrics across age gaps and appearance variations
- Document verification system with forgery detection capabilities
- Edge case handling protocols and human review escalation procedures

2.2.5 Abrar Zahin Raihan: User Experience, Fairness, and System Integration Lead

Primary Responsibilities:

- Design user-friendly verification interface with clear instructions in Bengali and English
- Implement fairness testing and bias mitigation across demographic groups
- Optimize decision thresholds to balance security with user experience
- Develop comprehensive monitoring and logging infrastructure
- Integrate all components into cohesive system with proper error handling

Problems to Solve:

- Precision vs. User Experience Trade-off: Balancing high fraud detection accuracy with low false positives to avoid blocking legitimate users
- Ensuring algorithmic fairness across diverse Bangladeshi populations (skin tones, ages, genders)
- Designing intuitive interfaces accessible to users with varying digital literacy levels
- Implementing graceful failure handling and helpful user guidance during verification issues

Key Deliverables:

- User interface design with accessibility features and multilingual support
- Threshold optimization analysis balancing false positives and false negatives
- System integration framework connecting all components
- Monitoring dashboard for tracking performance and bias metrics

2.3 Shared Responsibilities

All team members contribute to:

- Dataset collection and annotation for training and testing
- Documentation of technical decisions and system architecture
- Security review and vulnerability assessment
- Compliance with legal and ethical guidelines
- Performance evaluation and benchmarking
- User testing and feedback incorporation
- Code review and quality assurance

3 Background and Market Study

The rapid proliferation of digital financial services in emerging markets like Bangladesh has necessitated a shift from traditional Knowledge-Based Authentication (KBA) to robust biometric verification . However, the biometric landscape is currently under siege by a "black market" for identity replacement techniques, ranging from simple 2D printed photos to hyper-realistic 3D silicone masks and AI-generated deepfakes.

3.1 Current Threat Landscape

Modern fraud attempts are categorized into Presentation Attacks (PA), where physical artifacts like masks are used, and digital injection attacks, where synthetic media is fed directly into the system's data stream. The emergence of generative AI (GANs and diffusion models) has made high-fidelity spoofs accessible and scalable, creating an AI arms race where traditional movement-based liveness checks (e.g., blinking) are no longer sufficient as deepfakes can now convincingly mimic these actions .

3.2 Industrial Standards and Regulatory Framework

The global benchmark for evaluating these systems is the ISO/IEC 30107-3 standard for Presentation Attack Detection (PAD). Compliance levels range from Level 1 (basic 2D attacks) to Level 3 (expert-grade resin masks costing thousands of dollars). In the local context, Pathao Pay must adhere to Bangladesh Bank's regulations, which mandate secure KYC procedures involving valid National IDs (NID) and live in-app selfies.

3.3 Market Study: Leading Industrial Solutions

The commercial market is dominated by providers offering high-assurance, certified solutions:

- **FaceTec:** Industry leader in 3D FaceMaps, utilizing a \$600,000 spoof bounty to continuously harden their AI against evolving threats .
- **BioID:** A pioneer in motion-analysis algorithms that distinguish between 2D and 3D objects with a standard camera .
- **Yoti:** The first company to pass iBeta Level 3 PAD testing for passive, single-selfie liveness detection, establishing a new global benchmark for security .
- **HID Global:** Uses multi-spectral imaging to capture human tissue patterns across different light wavelengths, effectively seeing through artificial materials.

4 Overview of Solution Strategy and Technical Methods

4.1 Solution Architecture

- The proposed system employs a **multi-layered defense framework** with three tiers that progressively escalate suspicious cases, balancing security with real-time performance.

4.1.1 Three-Tier Detection Framework

- **Tier 1: On-Device Mobile Detection**
 - Lightweight CNNs (e.g., MobileNetV2/V3) using depthwise separable convolutions and inverted residual blocks.
 - Target efficiency: ~62 MB RAM with sub-25 ms latency.
 - Fine-tuning pre-trained ImageNet models on face anti-spoofing datasets.
 - Learned feature hierarchy:
 - * Early layers: edges and textures.
 - * Middle layers: facial structure consistency.
 - * Deep layers: high-level cues distinguishing natural vs. synthetic faces.
 - Known limitations and mitigations:
 - * Low-resolution cameras / poor lighting / long camera distance degrade performance.
 - * Quality-control fallbacks:
 - Prompt users to move closer if face is too small ($< 30\%$ of the frame).
 - Reject images below 480p.
 - Provide user guidance for improved lighting.
 - Escalate low-quality or ambiguous cases to server-side analysis.
- **Tier 2: Server-Side Deep Analysis**
 - Enables heavier models/analysis where computation constraints are relaxed.
 - Candidate paradigms to explore:
 - * Spatial-frequency hybrid learning.
 - * Multi-scale analysis (detect resolution-inconsistent artifacts).
 - * Domain generalization techniques.
 - * Generative reconstruction + residual analysis.
 - * Attention mechanisms focusing on telltale regions (e.g., eyes, skin texture).
 - Explored approaches so far:
 - * Semantic Anchor (unified physical-digital attack detection).
 - * OrthoDecomp (domain-invariant feature extraction).
 - * DeFakeGen (generative reconstruction with residual analysis).

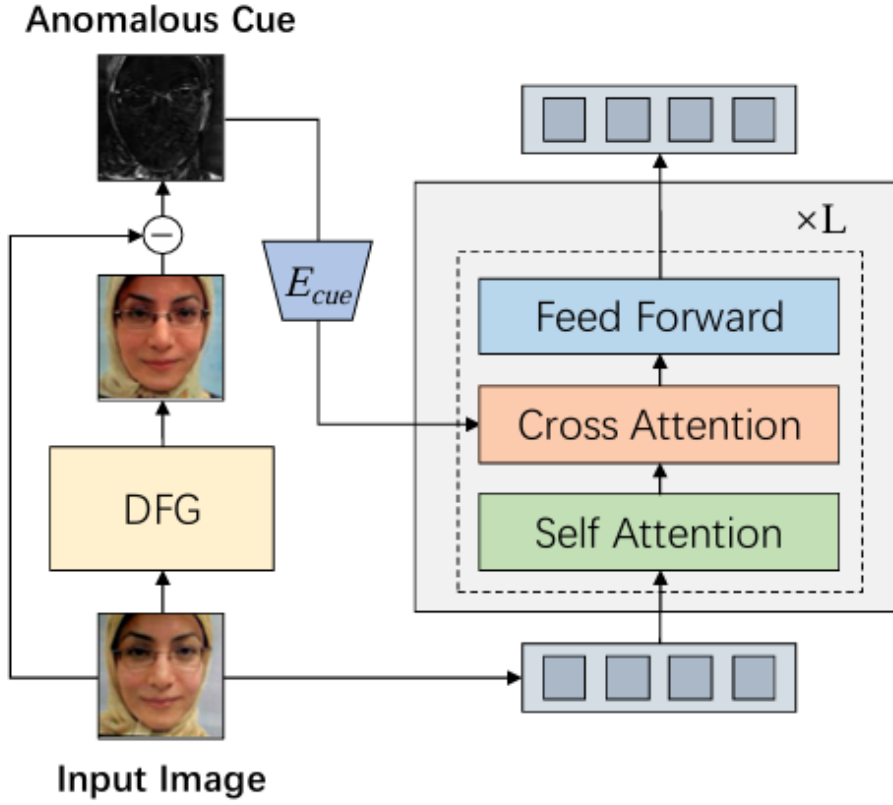


Figure 1: De-fake Face Generator (from *Generalized Face Liveness Detection via De-fake Face Generator*).

- Likely final strategy: **ensemble learning** (weighted voting / confidence fusion) to combine strengths across:
 - * GAN detection,
 - * diffusion-model signatures,
 - * presentation attack detection.

- **Tier 3: Human Review**

- Escalate low-confidence or conflicting predictions (expected ~2–5%) to trained analysts.
- Acts as a safety net and provides high-value labels for continuous improvement.

4.2 Training Data Strategy

4.2.1 Expected Industry Partnership Data

- Collaboration with Pathao Pay can provide:
 - real user selfies,
 - real attack attempts,

- device-specific metadata,
- geographic/demographic diversity representative of Bangladesh.
- This addresses critical gaps in public datasets (domain realism, device diversity, and population coverage).

4.2.2 Public Benchmark Datasets & Critical Gaps

- Supplementary public datasets:
 - CASIA-FASD,
 - OULU-NPU,
 - Idiap Replay-Attack,
 - CelebA-Spoof.
 - LCC-FASD
- Common limitations:
 - **Limited modern generative AI coverage:** StyleGAN3, Stable Diffusion, Mid-journey.
 - **Insufficient device variability:** missing budget smartphone sensor diversity.
 - **Demographic imbalance:** Western-centric; weak South Asian representation.
 - **Absence of hybrid attacks:** combined physical–digital manipulation.
- Gap mitigation plan:
 - generate synthetic samples using state-of-the-art generative models,
 - collect diverse real selfies from Pathao users,
 - simulate varied capture conditions,
 - create hybrid-attack samples.

4.3 Technical Implementation

4.3.1 Model Training Pipeline

- Three-stage approach:
 1. Baseline development on public datasets.
 2. Domain adaptation with Pathao data + continuous learning for new attack patterns.
 3. Ensemble optimization to minimize false positives while maintaining high detection rates.

4.3.2 Feature Analysis

- Beyond automatic CNN feature learning, explore explicit features such as:
 - frequency-domain cues for periodic patterns,
 - noise residual analysis for camera sensor inconsistencies,
 - biological signal detection where applicable.

4.3.3 Integration Architecture

- RESTful API workflow:
 - user captures selfie with real-time guidance,
 - on-device assessment,
 - quality gating with user feedback,
 - server escalation for flagged cases,
 - decision return with confidence score,
 - comprehensive logging for monitoring and retraining.

4.3.4 Evaluation Framework

- Standards-based evaluation: ISO/IEC 30107-3 PAD.
- Target metrics:
 - ACER $< 5\%$,
 - EER $< 3\%$,
 - ROC-AUC > 0.98 ,
 - on-device inference < 50 ms,
 - server analysis < 500 ms.
- Generalization testing:
 - cross-dataset evaluation to measure robustness to unseen attack types.

4.4 Future Extensions

- Modular upgrades supported by the architecture:
 - multimodal verification (e.g., voice pattern analysis),
 - active liveness cues (challenge–response movements),
 - NID cross-verification (matching against national ID photos),
 - adversarial robustness testing.
- Goal: evolve with generative AI advancements while minimizing friction for legitimate users.

5 Comparison of Existing Solutions vs. Proposed Solution

The proposed solution for Pathao Pay aims to bridge the gap between high-security commercial SDKs and flexible, cost-effective open-source frameworks, with a specific focus on the unique constraints of the Bangladeshi mobile market.

5.1 Comparative Analysis

Feature	Commercial Solutions (e.g., FaceTec, BioID)	Proposed Solution (Pathao Pay Optimization)
Accuracy	High (99%+); iBeta Level 2/3 certified .	Target ACER < 5% using ensemble deep learning.
Deployment	Often SaaS/Cloud-heavy or expensive on-premise SDKs.	Hybrid edge-cloud for offline-first resilience.
Hardware Requirements	Often requires high-end cameras or specific sensors .	Optimized for low-end Android devices (budget sensors) .
UX Friction	Mix of active and passive; can be high friction .	3-Tier framework: On-device screening → server deep analysis → human review.
Data Privacy	Managed by provider; potential "honeypot" risks.	On-device processing with immediate deletion of raw data .
Cost	High licensing fees; often per-transaction .	Cost-optimized via Vast.ai GPU training and open-source backbones .

Table 1: Comparison of current industrial leaders versus the proposed project architecture.

5.2 The Competitive Edge: Strategic Differentiation

While current market leaders provide exceptional security, they often struggle with the "last mile" of deployment in emerging economies. The proposed solution addresses three critical gaps:

1. **Demographic and Environmental Adaptation:** Unlike many Western-trained models, our solution will utilize domain adaptation to handle South Asian skin tones and inconsistent lighting conditions characteristic of the local environment .

2. **Low-End Device Optimization:** By utilizing lightweight backbones like MobileNetV3 and knowledge distillation, we ensure real-time performance on devices common in Bangladesh that lack GPU acceleration .
3. **Adversarial Defense:** We incorporate explicit anomalous cue detection (e.g., DeFakeGen residual analysis) to identify generative artifacts that traditional commercial sensors might miss if they rely solely on depth perception .

6 Analysis of Economic and Environmental Impact

6.1 Economic Impact

6.1.1 Fraud Prevention and Direct Cost Savings

- **Financial Loss Mitigation:** Effective liveness detection prevents unauthorized account access and identity theft in mobile financial transactions. According to industry estimates, financial fraud in digital payment systems can account for 0.5-2% of transaction volumes; robust verification can reduce these losses significantly.
- **Operational Efficiency:** Automated verification reduces manual KYC processing costs. The three-tier architecture (on-device → server → human review at 2-5%) minimizes expensive human review requirements while maintaining security standards.
- **Regulatory Compliance:** Meeting Bangladesh Bank’s KYC requirements avoids penalties and maintains operating licenses, protecting Pathao Pay’s ability to conduct business.

6.1.2 Market and Business Impact

- **User Trust and Growth:** Enhanced security builds consumer confidence in digital financial services, potentially accelerating adoption in Bangladesh’s growing mobile payment market.
- **Competitive Positioning:** Advanced fraud detection capabilities differentiate Pathao Pay from competitors in the crowded mobile financial services sector.
- **Cost-Effective Implementation:** Project budget of \$260-540 USD using Vast.ai GPU training represents significant savings compared to commercial SDK licensing fees (often \$10,000-100,000+ annually) or per-transaction costs.

6.1.3 Financial Inclusion Considerations

- **Accessibility:** Mobile-first optimization (62 MB RAM, sub-3 second latency) enables verification on low-end Android devices common among economically disadvantaged populations, expanding financial service access rather than restricting it.
- **User Friction Management:** Balanced false positive rates prevent legitimate users from being excluded from financial services, which is critical for maintaining inclusive access while ensuring security.

6.2 Environmental Impact

6.2.1 Computational Resources and Energy Consumption

- **Training Phase Energy Use:** The project requires 700-900 GPU hours for model training on RTX 4090/5090 GPUs. Modern high-performance GPUs consume approximately 300-450 watts during intensive training. Total estimated training energy: 210-405 kWh over the 6-month development period.

- **Inference Energy Efficiency:** On-device processing using lightweight MobileNet models (targeting <50ms inference) consumes minimal power compared to server-based processing, reducing ongoing operational energy requirements and carbon footprint.
- **Cloud Infrastructure:** Server-side Tier 2 analysis will require cloud computing resources. However, the three-tier design ensures only flagged cases (estimated 10-20% of submissions) require server processing, minimizing unnecessary cloud computation.

6.2.2 Sustainability Strategies

- **Edge Computing Priority:** Performing initial screening on-device reduces data transmission and cloud processing requirements, lowering network energy consumption and server load.
- **Model Efficiency:** Knowledge distillation and quantization techniques reduce model size and computational requirements, enabling energy-efficient inference on mobile CPUs without GPU acceleration.
- **Resource Optimization:** Using Vast.ai spot instances allows training on existing GPU capacity that would otherwise be idle, representing more efficient resource utilization than dedicated infrastructure.

6.2.3 Long-Term Environmental Considerations

- **Hardware Lifecycle:** The system’s compatibility with existing low-end mobile devices reduces the need for hardware upgrades, extending device lifecycles and reducing electronic waste.
- **Scalability Impact:** As user base grows, the on-device processing architecture prevents linear scaling of cloud infrastructure energy requirements, as most verifications complete locally.
- **Continuous Training:** Ongoing model updates will require periodic retraining (estimated quarterly), adding 70-90 GPU hours per year for maintenance, representing modest ongoing environmental impact.

6.3 Net Impact Assessment

The economic benefits of fraud prevention and financial inclusion substantially outweigh implementation costs. The environmental footprint is modest compared to alternative approaches requiring constant cloud processing, and the edge-first architecture demonstrates consideration for sustainable computing practices. The system’s design prioritizes resource efficiency without compromising security effectiveness, aligning economic viability with environmental responsibility.

7 Legal, Safety, Social, and Ethical Considerations

The implementation of AI-powered liveness detection in Pathao Pay raises critical legal, ethical, and social concerns that must be addressed to ensure responsible deployment. As a mobile financial service operating in Bangladesh, the system must balance fraud prevention with user rights, financial inclusion, and cultural appropriateness while complying with national regulations and international best practices.

7.1 Legal and Regulatory Compliance

7.1.1 Data Protection and Privacy

The collection and processing of biometric data must comply with Bangladesh's legal framework:

- **National Regulations:** Adherence to the ICT Act 2006 and any applicable regulations governing biometric data collection and processing in mobile financial services
- **Biometric Data Sensitivity:** Recognition that facial biometric data constitutes highly sensitive personal information requiring explicit user consent, enhanced security measures, and strict purpose limitation
- **Cross-Border Data Transfer:** Compliance with regulations if biometric data is transmitted internationally for cloud processing or model training
- **User Rights Implementation:** Providing mechanisms for users to access, correct, and delete their biometric data, along with transparent privacy policies in both Bengali and English

7.1.2 Financial Services Regulations

The system must satisfy Bangladesh Bank's requirements for digital financial services:

- **KYC Compliance:** Meeting Know Your Customer standards for customer identification and verification in mobile financial services
- **Audit Requirements:** Maintaining comprehensive logs of verification attempts and fraud detection for regulatory examination and compliance reporting
- **Consumer Protection:** Ensuring verification processes comply with consumer protection regulations and do not unfairly exclude legitimate users

7.1.3 Liability and Accountability

Clear accountability frameworks must address potential failures:

- Responsibility for false rejections that deny legitimate users access to financial services
- Liability when fraudulent actors successfully bypass verification
- Data breach notification obligations and remediation requirements
- Mechanisms for users to dispute automated verification decisions

7.2 Ethical Considerations

7.2.1 Algorithmic Fairness and Bias

Facial recognition systems have demonstrated concerning disparities that must be actively mitigated:

- **Demographic Representation:** Ensuring training datasets adequately represent Bangladesh’s diverse population, including various skin tones, ethnicities, age groups, and gender presentations
- **Performance Equity:** Conducting rigorous fairness audits to verify consistent accuracy across demographic segments, with particular attention to historically marginalized groups
- **Bias Monitoring:** Implementing continuous performance monitoring in production to detect and address emerging biases
- **Testing with Local Populations:** Validating system performance specifically with Bangladeshi users rather than relying solely on international datasets

7.2.2 Financial Inclusion

The system must enhance rather than hinder access to financial services:

- **Device Accessibility:** Supporting low-end Android devices commonly used by economically disadvantaged populations who most need mobile financial services
- **Digital Literacy:** Designing intuitive interfaces accessible to users with limited technical experience, with clear Bengali language instructions
- **Network Constraints:** Functioning reliably in areas with poor connectivity or limited bandwidth
- **Alternative Pathways:** Providing fallback verification methods for users with disabilities, facial differences, or circumstances preventing standard liveness detection
- **Documentation Flexibility:** Accommodating users with various forms of identification beyond standard NID cards

7.2.3 Transparency and User Autonomy

Users deserve clear understanding and control over biometric verification:

- **Informed Consent:** Obtaining explicit consent with clear explanations in accessible language about biometric data collection, usage, and retention
- **Decision Explanation:** Providing meaningful feedback when verification fails, helping users understand issues without revealing exploitable security details
- **Human Review:** Offering pathways for human oversight in disputed cases rather than relying solely on automated decisions
- **System Limitations:** Honestly communicating potential error rates and circumstances where verification may be unreliable

7.3 Safety and Security Considerations

7.3.1 Biometric Data Security

The irreversible nature of facial biometrics demands exceptional protection:

- **Encryption:** End-to-end encryption for data transmission and encryption-at-rest for stored biometric templates
- **Data Minimization:** Processing biometric data on-device where possible and storing only mathematical templates rather than raw images
- **Access Controls:** Strict limitations on personnel access to biometric databases
- **Secure Deletion:** Cryptographic destruction of biometric data when users close accounts or withdraw consent

7.3.2 Misuse Prevention

Technical and organizational safeguards must prevent harmful applications:

- **Purpose Limitation:** Ensuring biometric data is used exclusively for identity verification and not repurposed for marketing, surveillance, or profiling
- **Third-Party Restrictions:** Prohibiting sale or sharing of biometric data with external parties
- **Government Access Protocols:** Establishing clear legal standards and judicial oversight for any law enforcement access requests

7.4 Social and Cultural Considerations

The system must respect Bangladesh's cultural context:

- **Religious Accommodations:** Providing alternative verification methods for users who wear religious face coverings (hijabs, niqabs) in accordance with Islamic modesty practices
- **Gender Considerations:** Ensuring reliable performance for users across gender presentations, including transgender and gender non-conforming individuals
- **Privacy Norms:** Respecting cultural attitudes toward facial photography and biometric data collection
- **Language Accessibility:** Offering complete functionality in Bengali with culturally appropriate communication styles

7.5 Design Principles for Responsible Implementation

The following principles should guide system development and deployment:

- **Privacy by Design:** Embedding privacy protections from inception, defaulting to minimal data collection and maximum user control
- **Inclusive Design:** Centering marginalized users in design processes and testing with diverse Bangladeshi populations
- **Proportionality:** Ensuring security measures are justified by genuine fraud risks rather than implementing maximum restrictions by default
- **Reversibility:** Maintaining options to transition away from biometric verification if deemed harmful or unnecessary
- **Continuous Improvement:** Regularly reviewing and updating systems based on user feedback, fairness audits, and emerging best practices

8 Timeline and Milestones

The project lifecycle is strategically divided into research, specialized system design for adversarial environments, and the development of a resource-optimized prototype for the Bangladeshi mobile market.

8.1 Project Milestones

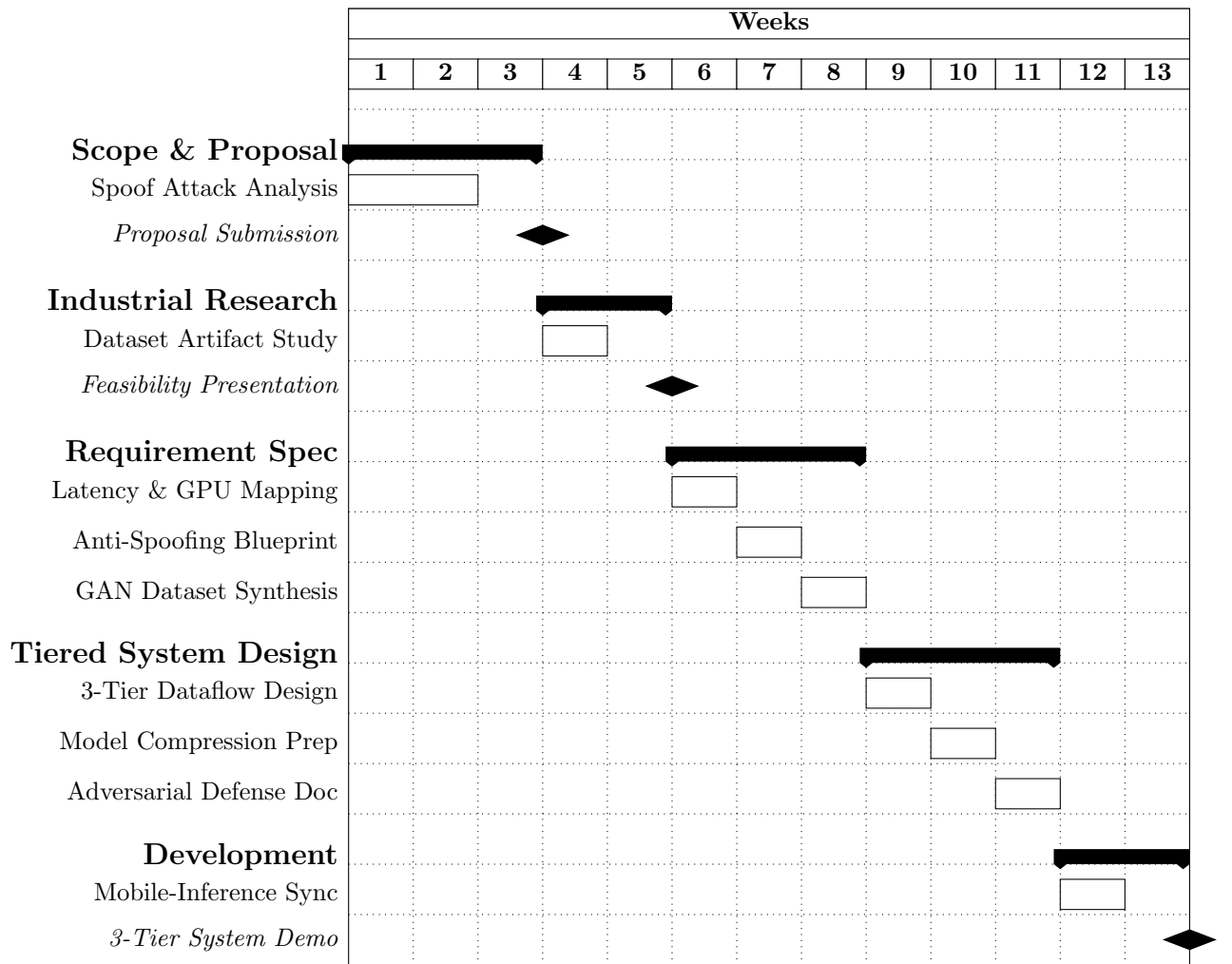
These milestones serve as validation gates ensuring that technical objectives—from dataset synthesis to mobile-edge deployment—are achieved according to the security requirements of Pathao Pay.

Milestone	Specific Technical Activity / Deliverable	Week	COs
M1	Domain Research: Bio-metric Presentation Attack Detection (PAD)	Week 1	-
M2	CEP Identification: Identifying GAN vs. Diffusion Spoofing vectors	Week 2	-
M3	PROPOSAL: Secure Liveness & NID Match Scope for Pathao	Week 3	CO1, 5, 7
M4	Industrial Benchmarking: Analysis of FaceTec & Yoti architectures	Week 4	-
M5	PRESENTATION: Robustness & Fairness (Bias) Study	Week 5	CO1-5, 7
M6	Engineering: Sensor-specific latency constraints for budget Androids	Week 6	-
M7	INCEPTION REPORT: Multi-layered Anti-Spoofing Blueprint	Week 7	CO1-8
M8	Data Policy: Synthesis of GAN-spoof datasets	Week 8	-
M9	PRESENTATION: 3-Tier Logic (On-device vs Server-side)	Week 9	CO1, 3, 4
M10	Design: MobileNet Quantization & Flask Backend Integration	Week 10	-
M11	REPORT: System Analysis & Adversarial Defense Design	Week 11	CO3, 4
M12	Prototyping: Integrating CNN Inference with Mobile Camera	Week 12	-
M13	DEMO: 3-Tier Liveness System (Mobile Detection + Server Validation)	Week 13	CO8

Table 2: Project-Specific Milestones and Alignment with Course Outcomes.

8.2 Timeline

The following chart outlines the specialized workstreams: Researching deepfake artifacts (Phase 2), designing the tiered dataflow (Phase 4), and optimizing for mobile-edge performance (Phase 5).



9 Budget Plan and Estimated Costs

9.1 Overview

This budget supports a 6-month development cycle for a multi-stage detection pipeline, leveraging cost-effective GPU compute via Vast.ai for training both lightweight on-device models (MobileNet) and heavier server-side deep analysis models while minimizing costs through academic resources and open-source tools.

9.2 Cost Breakdown

9.2.1 Cloud Computing and GPU Resources

- **GPU Training via Vast.ai**

- Platform: Vast.ai (significantly cheaper than GCP/AWS)
- Configuration: RTX 4090/5090 or similar high-performance GPUs (\$0.20-0.40/hour)
- Usage breakdown:
 - * Tier 1 - MobileNet optimization: 150-200 GPU hours
 - * Tier 2 - Heavy server models (ensemble, DeFakeGen, etc.): 400-500 GPU hours
 - * Adversarial training and robustness testing: 150-200 GPU hours
- Total: 700-900 GPU hours
- Cost: \$140-360

- **Cloud Storage:** Google Drive/Dropbox free tiers or university storage

- **Development & Testing:** Local machines + free cloud tiers

Subtotal: \$140-360

9.2.2 Software and Services

- Development tools: Free (PyTorch, TensorFlow, OpenCV)
- Datasets: Free (public datasets)
- Annotation: Manual (team effort)
- Version control: Free (GitHub student)
- Mobile testing: Personal devices

Subtotal: \$0

9.2.3 Contingency

- Additional Vast.ai credits: \$80-120
- Unexpected costs: \$40-60

Subtotal: \$120-180

9.3 Total Budget

Category	Min (USD)	Max (USD)
Cloud & GPU (Vast.ai)	140	360
Contingency	120	180
Total	260	540

9.4 Cost Optimization Strategy

- **Vast.ai:** Use spot instances at \$0.20-0.40/hr vs. \$1-3/hr on major clouds
- **Multi-Stage Efficiency:** Train lightweight MobileNet first, then focus GPU resources on complex server-side models
- **University Resources:** Campus GPU clusters for initial experiments
- **Free Storage:** Google Drive, Dropbox, or university storage systems
- **Team Devices:** Personal smartphones for mobile testing
- **Open Source:** All development tools are free
- **Transfer Learning:** Use pretrained models to reduce training time for both tiers

Realistic Budget: \$260-540 USD